1.6 Sums of Two Squares

WORKSHEET

The question

STEP II 2023, Question 7

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself If $z = a + \mathrm{i}b$ with a and b integers, what is $|z|^2$?

STAGE 1 Moduli multiply

Covers: Part (i)

Pause and try Multiply out zw, write down $|zw|^2$, and expand. Can you factorise what you get?

STAGE 2 Using it

Covers: Part (ii)

Ask yourself Write 5 and 221 as sums of two squares. Which complex numbers have $|z|^2 = 5$ and $|w|^2 = 221$?

Pause and try By part (i), $|zw|^2 = 5 \times 221$. Multiply out zw: its real and imaginary parts are h and k. For a second answer, try the conjugate of one of the factors.

STAGE 3 Factorise first

Covers: Parts (iii) and (iv)

Ask yourself 8045 ends in 5. Factorise it.

Pause and try The other factor is just above a perfect square. How far above, and is that difference a square? Then multiply, as in part (ii).

Pause and try Part (iv): 36 is already a square. Which complex number has modulus squared 36, and what does multiplying by it do?

STAGE 4 Three ways

Covers: Part (v)

Ask yourself As in part (iii), the large number is close to a perfect square. Write it as a sum of two squares first.

Pause and try 25 can be written as $|\alpha|^2$ in more than one way: 5 is one. How many Gaussian integers have $|\alpha|^2 = 25$? Which of them give genuinely different pairs?

STAGE 5 Dividing

Covers: Part (vi)

Ask yourself Write 1002037 and 109 as sums of two squares; look for a nearby perfect square again.

Pause and try Parts (ii) to (v) multiplied. This part runs the idea backwards: divide one Gaussian integer by the other. If the division does not come out exactly, try the conjugate.

What's really going on

Fermat's two-squares theorem

Which primes are sums of two squares? $2 = 1^2 + 1^2$, 5, 13, 17, 29, 37, 41 are; 3, 7, 11, 19, 23, 31 are not. The pattern: apart from 2, a prime is a sum of two squares exactly when it leaves remainder 1 on division by 4. Fermat stated this in 1640; Euler proved it in 1749. In the language of Gaussian integers it says: a prime $p \equiv 1 \pmod 4$ *splits* as $p = (a + bi)(a - bi)$, while a prime $p \equiv 3 \pmod 4$ stays prime. For example, $5 = (2 + i)(2 - i)$, but 3 cannot be factorised at all.

In 1990 Don Zagier published a proof of the two-squares theorem that is famously one sentence long. It appears, with explanation, in *Proofs from THE BOOK*.

A new number system with its own primes

Gauss realised that the Gaussian integers behave remarkably like the ordinary integers. There is a division algorithm and a Euclidean algorithm, and every Gaussian integer factorises *uniquely* into Gaussian primes (up to multiplying by ± 1 or $\pm i$). Combined with $(\star)$, this lets you count exactly how many ways a number can be written as a sum of two squares, just from its prime factorisation.

At university this is the gateway to *algebraic number theory*: studying number systems such as $\mathbb{Z}[i]$ or $\mathbb{Z}[\sqrt{-5}]$ in order to understand ordinary whole numbers. Section 1.7 shows why the subject needed to be invented: not every such system is as well behaved as the Gaussian integers.

Four squares and beyond

Every positive integer is a sum of *four* squares (Lagrange, 1770). The proof relies on a four-square version of $(\star)$, which comes from Hamilton's *quaternions* in the same way that $(\star)$ comes from complex numbers. You will see it in Section 7.8.

Try this next

1. **An obstruction.** Show that no number of the form $4k + 3$ is a sum of two squares. (What are the squares modulo 4?)

2. **Two ways.** Write 65 and 85 each as a sum of two squares in two different ways, and explain your answers using Gaussian integers.

3. **Four ways.** $1105 = 5 \times 13 \times 17$. Find four different ways to write 1105 as a sum of two positive squares. Why four?

4. **Gaussian primes.** HARDER Show that if $3 = (a + bi)(c + di)$ with a, b, c, d integers, then one of the factors has modulus 1. (Take moduli squared.) This shows that 3 is a Gaussian prime.

Further reading

- **John Stillwell, *Elements of Number Theory*** (Springer). Builds up to Gaussian integers and beyond, with history.
- **Aigner & Ziegler, *Proofs from THE BOOK*,** Chapter 4, "Representing numbers as sums of two squares".
- **Joseph Silverman, *A Friendly Introduction to Number Theory*,** chapters on sums of squares and Gaussian integers.

The full guided solution is in Section 1.6:
https://beyondstep.keithmcnulty.org/ch01/two-squares.html